

LLM Alignment

Why Go Beyond RLHF?

- ▶ RLHF was a major breakthrough for aligning LLMs with human preferences.
- ▶ But the standard pipeline is relatively complex:
 - ▶ Step 1: Start from a supervised fine-tuned (SFT) model.
 - ▶ Step 2: Train a separate reward model from human preference data.
 - ▶ Step 3: Use reinforcement learning (typically PPO) to optimize the policy.
- ▶ This pipeline has a lot of moving pieces and is hard to get right in practice:
 - ▶ you need to train a separate reward model,
 - ▶ you have an RL optimization loop,
 - ▶ you are dealing with potential instability during training.
- ▶ Why not learn directly from preferences?

Direct Preference Optimization (DPO)¹

- ▶ DPO asks:
 - ▶ Do we really need a separate reward model and PPO stage?
- ▶ DPO shows that we can train the model directly on preference pairs.
- ▶ Each training example contains:
 - ▶ a prompt x ,
 - ▶ a preferred response y_w ,
 - ▶ and a rejected response y_l .
- ▶ The goal is simple:
 - ▶ increase the probability of the preferred response,
 - ▶ decrease the probability of the rejected response.
- ▶ So DPO turns alignment into a supervised-style optimization problem rather than a full RL pipeline.

¹Rafael Rafailov et al. 2023. *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*. NeurIPS 2023.

The Key Insight Behind DPO

- ▶ In classical RLHF, the policy is optimized to:
 - ▶ achieve high reward,
 - ▶ while staying close to a reference policy.
- ▶ DPO shows that this KL-constrained objective has a useful implication:
 - ▶ the reward can be expressed in terms of how much the policy prefers an output relative to the reference model.
- ▶ In that sense, the language model is *secretly* acting like a reward model.
 - ▶ There is a mapping between reward functions and optimal policies under the KL-constrained objective, which lets us optimize preferences directly
 - ▶ Instead of explicitly learning reward first and then doing RL, we can optimize preferences directly.

What DPO Optimizes

- ▶ DPO uses pairs of responses for the same prompt:
 - ▶ one chosen by humans,
 - ▶ one rejected by humans.
- ▶ The model is trained to prefer the chosen response over the rejected one.
- ▶ This preference is measured relative to a reference model, usually the original supervised fine-tuned model before DPO training.
- ▶ Intuition:
 - ▶ make the preferred answer more likely;
 - ▶ make the rejected answer less likely;
 - ▶ stay close to the reference model.
- ▶ So the reference model acts as an anchor: DPO improves preferences while keeping the learned policy close to the original model.

DPO vs PPO

- ▶ DPO simplifies preference-based alignment.
- ▶ Compared with classical RLHF, DPO:
 - ▶ avoids training a separate reward model
 - ▶ avoids a PPO optimization loop
 - ▶ uses a classification-style loss
- ▶ This makes training:
 - ▶ easier to implement
 - ▶ more stable in practice
 - ▶ more computationally efficient

The big picture

- ▶ A language model assigns probabilities to possible completions of text.
- ▶ Before alignment, those probabilities reflect patterns learned from large text datasets, not necessarily what humans prefer most.
- ▶ The goal of alignment is to shift probability mass toward responses that people judge to be better.
- ▶ In the DPO derivation, the reward signal can be written in terms of the policy's log-probabilities relative to a reference model.
- ▶ This lets us train directly on human preference comparisons, without separately training a standalone reward model.

Starting Point for the DPO Derivation

$$\max_{\pi} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(y|x)} [r(x, y)] - \beta D_{\text{KL}} [\pi(y | x) \parallel \pi_{\text{ref}}(y | x)]$$

- ▶ This is the KL-regularized RLHF objective over the policy π .
- ▶ We use it as a *starting point* to derive DPO.
- ▶ DPO will eliminate the need to explicitly learn a separate reward model.
- ▶ The key result will relate reward differences to

$$\log \pi(y | x) - \log \pi_{\text{ref}}(y | x).$$

Step 1: Start from the KL-Regularized RLHF Objective

$$\max_{\pi} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(y|x)} [r(x, y)] - \beta D_{\text{KL}} [\pi(y | x) \parallel \pi_{\text{ref}}(y | x)]$$

- ▶ $x \sim \mathcal{D}$: prompts come from the dataset.
- ▶ $y \sim \pi(y | x)$: responses are sampled from the current policy.
- ▶ $r(x, y)$: reward assigned to response y for prompt x .
- ▶ $\pi_{\text{ref}}(y | x)$: the reference policy.
- ▶ β : controls the strength of the KL penalty.

Interpretation of the Objective

- ▶ The first term,

$$\mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(y|x)}[r(x, y)],$$

encourages the policy to generate high-reward responses.

- ▶ The second term,

$$\beta D_{\text{KL}}[\pi(y | x) \parallel \pi_{\text{ref}}(y | x)],$$

penalizes deviation from the reference policy.

- ▶ So the objective means:
 - ▶ maximize reward,
 - ▶ while staying close to $\pi_{\text{ref}}(y | x)$.

Step 2: Expand the KL Term

Using the definition of KL divergence,

$$D_{\text{KL}}[\pi(y | x) \| \pi_{\text{ref}}(y | x)] = \mathbb{E}_{y \sim \pi(y|x)} \left[\log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} \right],$$

the objective becomes

$$\max_{\pi} \mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y \sim \pi(y|x)} \left[r(x, y) - \beta \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} \right].$$

Step 3: Rewrite as a Minimization Problem

This is equivalent to

$$\min_{\pi} \mathbb{E}_{x \sim \mathcal{D}} \mathbb{E}_{y \sim \pi(y|x)} \left[\log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)} - \frac{1}{\beta} r(x, y) \right].$$

- ▶ This is just algebra:
- ▶ maximizing $A - B$ is the same as minimizing $\frac{B-A}{\beta}$.
- ▶ Nothing conceptual has changed yet.

Step 4: Introduce the Partition Function

Rewrite the inner term using $-C = -\log e^C$:

$$\begin{aligned}\log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} - \frac{1}{\beta} r(x, y) &= \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} - \log \exp\left(\frac{1}{\beta} r(x, y)\right) \\ &= \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x) \exp\left(\frac{1}{\beta} r(x, y)\right)}.\end{aligned}$$

For each prompt x , we define a normalizing constant called partition function

$$Z(x) = \sum_y \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right).$$

Think of $Z(x)$ as summing over every possible completion for prompt x

$$\begin{aligned} \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right)} &= \log \frac{\pi(y \mid x)}{\frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right)} \\ &\quad - \log Z(x). \end{aligned}$$

This turns the denominator into a normalized probability distribution.

Step 5: Define the Target Distribution π^*

Define

$$\pi^*(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right).$$

► $\pi^*(y \mid x) \geq 0$ for all y .

► By construction,

$$\sum_y \pi^*(y \mid x) = 1.$$

► So $\pi^*(y \mid x)$ is a valid probability distribution.

Interpretation of $\pi^*(y \mid x)$

$$\pi^*(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

- ▶ Start from the reference policy $\pi_{\text{ref}}(y \mid x)$.
- ▶ Reweight each response y by

$$\exp\left(\frac{1}{\beta} r(x, y)\right).$$

- ▶ So:
 - ▶ higher reward $\Rightarrow$ more probability mass,
 - ▶ lower reward $\Rightarrow$ less probability mass.
- ▶ This is the reward-tilted version of the reference policy.

Step 6: Rewrite the Objective Using π^*

Substituting $\pi^*(y | x)$ into the objective gives

$$\min_{\pi} \mathbb{E}_{x \sim \mathcal{D}} \left[\mathbb{E}_{y \sim \pi(y|x)} \log \frac{\pi(y | x)}{\pi^*(y | x)} - \log Z(x) \right]$$

The inner expectation is exactly

$$D_{\text{KL}}[\pi(y | x) \parallel \pi^*(y | x)].$$

So the objective becomes

$$\min_{\pi} \mathbb{E}_{x \sim \mathcal{D}} \left[D_{\text{KL}}[\pi(y | x) \parallel \pi^*(y | x)] - \log Z(x) \right]$$

Why the $Z(x)$ Term Does Not Affect the Optimum

- ▶ The term

$$-\log Z(x)$$

depends on x , the reward function, and the reference policy.

- ▶ It does not depend on the policy π .
- ▶ Therefore, when optimizing with respect to π , it is just an additive constant.
- ▶ So the optimization is equivalent to minimizing

$$D_{\text{KL}}[\pi(y | x) \| \pi^*(y | x)].$$

Step 7: Use the Nonnegativity of KL Divergence

We now use the fact that

$$D_{\text{KL}}(P \| Q) \geq 0$$

with equality if and only if $P = Q$.

- Therefore, the minimum is achieved exactly when

$$\pi(y \mid x) = \pi^*(y \mid x)$$

for all x .

- So the optimal policy is

$$\pi(y \mid x) = \pi^*(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

Closed-form optimal policy

$$\pi^*(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right)$$

- ▶ This is the closed-form optimal policy under the KL-regularized RLHF objective.
- ▶ It says:
 - ▶ the optimal policy is the reference policy,
 - ▶ exponentially tilted toward higher-reward responses.
- ▶ Mathematically:
 - ▶ Start with the reference LLM π_{ref}
 - ▶ Upgrade the responses by multiplying the probability of things that humans like by the exponentiated reward $\exp\left(\frac{1}{\beta} r(x, y)\right)$
 - ▶ Renormalize for each prompt x using the partition function, i.e., a prompt-dependent normalization function $Z(x)$

This is the key result that DPO builds on.

Where are we at?

- ▶ We have a closed-form expression for the optimal policy that gives a relationship between the optimal policy and the reward model
- ▶ This result was known before DPO.
- ▶ But by itself, it is not directly usable in practice.
 - ▶ Why? Because it depends on the normalization term $Z(x)$.
 - ▶ Computing $Z(x)$ is intractable.

Step 8: Rearranging the Optimal Policy Formula

Take logs of

$$\pi^*(y \mid x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right).$$

Then

$$\log \pi^*(y \mid x) = \log \pi_{\text{ref}}(y \mid x) + \frac{1}{\beta} r(x, y) - \log Z(x).$$

Rearranging gives

$$r(x, y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x).$$

If the optimal policy π^* puts more (less) probability on a response y to a prompt x compared to π_{ref} , the reward is higher (lower).

Are we there yet?

- ▶ No, because we still have that pesky partition function $Z(x)$
- ▶ To make it go away, think back about RLHF
 - ▶ Loss function on reward functions

$$\mathcal{L}_R(r, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} [\log \sigma(r(x, y_w) - r(x, y_l))]$$

- ▶ Transformation between reward functions and policies

$$r(x, y) = \beta \log \frac{\pi^*(y | x)}{\pi_{\text{ref}}(y | x)} + \beta \log Z(x).$$

- ▶ We can get a loss function on policies!

Step 9: Bring in the Bradley-Terry Preference Model

Assuming a Bradley-Terry preference model:

$$p^*(y_1 \succ y_2 \mid x) = \sigma(r^*(x, y_1) - r^*(x, y_2)).$$

- ▶ Preference probabilities depend only on reward differences.
- ▶ This will help us get rid of $Z(x)$, because the extra term

$$\beta \log Z(x)$$

depends only on x , not on y .

Step 10: The Partition Function Cancels

Using

$$r(x, y) = \beta \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} + \beta \log Z(x),$$

we get

$$r(x, y_1) - r(x, y_2) = \beta \log \frac{\pi(y_1 | x)}{\pi_{\text{ref}}(y_1 | x)} - \beta \log \frac{\pi(y_2 | x)}{\pi_{\text{ref}}(y_2 | x)}.$$

The terms

$$\beta \log Z(x) \quad \text{and} \quad \beta \log Z(x)$$

cancel because they are the same for both responses.

- This is the step that removes the need to compute the partition function explicitly.

Step 11: Preference Probability in Terms of the Policy

So the human preference probability can be written directly in terms of the policy:

$$p^*(y_w \succ y_l \mid x) = \sigma \left(\beta \log \frac{\pi^*(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi^*(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right).$$

- ▶ This is the bridge from reward modeling to direct policy optimization.
- ▶ We no longer need an explicit reward model as the object being trained.

Step 12: Parameterize the Policy as π_θ

Now replace the unknown optimal policy π^* with a parameterized policy π_θ .

Given preference triples

$$(x, y_w, y_l) \sim \mathcal{D},$$

we model

$$P(y_w \succ y_l \mid x) = \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right).$$

- ▶ y_w : preferred response
- ▶ y_l : rejected response

From Preference Probabilities to a Loss

From the previous slide, the model assigns probability

$$P(y_w \succ y_l \mid x) = \sigma \left(\beta \log \frac{\pi_{\theta}(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_{\theta}(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right)$$

to the observed preference pair.

- ▶ Each training example says that, for prompt x , y_w was preferred to y_l .
- ▶ So we want the model to assign high probability to that observed event.
- ▶ Maximum likelihood therefore tries to maximize

$$P(y_w \succ y_l \mid x)$$

over all preference pairs in the dataset.

- ▶ Equivalently, we minimize the negative log-likelihood:

$$-\log P(y_w \succ y_l \mid x).$$

Step 13: The DPO Objective

Applying negative log-likelihood to the preference model gives

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} [\log P(y_w \succ y_l \mid x)].$$

Substituting the DPO preference model,

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_{\theta}(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right]$$

- ▶ This is the final DPO objective
 - ▶ Minimizing loss = maximizing difference
 - ▶ Our policy should put a higher probability on preferred responses compared to the reference policy...
 - ▶ ... and a lower probability on dispreferred responses compared to π_{ref}

How to Read the DPO Loss

- ▶ The loss is small when the model assigns a much larger

$$\log \frac{\pi_{\theta}(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)}$$

to the preferred response than to the rejected one.

- ▶ So DPO encourages the policy to:
 - ▶ increase the relative probability of preferred responses,
 - ▶ decrease the relative probability of rejected responses,
 - ▶ while measuring everything relative to the reference model.
- ▶ This relative comparison is what plays the role of KL-style regularization.

Big Picture: What the Derivation Proved

1. Start from the KL-regularized RLHF objective.
2. Derive the closed-form optimal policy:

$$\pi(y \mid x) \propto \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right).$$

3. Rewrite reward differences using policy/reference log-ratios.
4. Use the Bradley-Terry preference model, where only reward differences matter.
5. The partition function cancels.
6. This yields a direct preference loss over π_{θ} : DPO.